

Chapter 17 – BSOD, Servers & Active Directory (AD)

1. BSOD (Blue/Black Screen of Death)

👉 In older Windows → **Blue Screen of Death (BSOD)**

👉 In newer OS → sometimes appears as **Black Screen of Death**

What is BSOD?

BSOD is a **critical system failure screen** that stops Windows immediately to prevent damage.

Real-Life Example:

👉 Like an **MCB (Mini Circuit Breaker)** in your house ⚡

- If too much current comes → MCB trips
 - Similarly → if system overload happens → BSOD appears
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Why BSOD Happens? (Very Important for Interview)

◆ 1. RAM Overload

Example:

- Total RAM = 16GB
- OS uses = 3.7GB
- Apps use = 14GB

👉 Total = 17.7GB ❌ (more than capacity)

👉 System retries → overload → **BSOD**

◆ 2. Faulty RAM

👉 Use Windows Memory Diagnostic tool:

- Checks if RAM is working properly

◆ 3. HDD / SSD Issues

- Bad sectors
 - Incorrect indexing
- 👉 Causes file corruption → BSOD

◆ 4. Driver Issues

- Old drivers
- Incompatible drivers

👉 New OS cannot handle old driver → **Kernel Error**

◆ 5. OS Corruption

- System32 files damaged
- Missing files

👉 OS fails → BSOD

◆ 6. Processor Overload

- Heavy tasks → CPU overload
- 👉 System crash

◆ 7. SMPS (Power Supply)

- Voltage fluctuation
- 👉 Sudden shutdown → BSOD

Quick Troubleshooting for BSOD

- ✓ Check RAM usage
- ✓ Run Memory Diagnostic
- ✓ Update drivers (official website only)
- ✓ Run:
 - sfc /scannow
 - chkdsk
 - ✓ Check power supply
 - ✓ Remove recent apps

2. What is a Server?

 A server provides services to client machines

Example:

- Your laptop = Client 
- Server = Service provider 

Types of Servers (Must Know)

Server Type	Function
 Web Server	Hosts websites
 App Server	Runs applications
 DNS Server	Converts domain → IP
 VPN Server	Secure remote access
 File Server	Stores & shares files
 Database Server	Stores structured data

Server Type	Function
 AD Server	Manages users & systems
 Mail Server	Handles emails
 VMware Server	Virtualization
 DHCP Server	Assigns IP automatically

3. Active Directory (AD) Deep Understanding

👉 Active Directory = **Centralized Database System**

Stores:

- Users 
 - Computers 
 - Printers 
 - Policies 
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Key Components

User

- Individual person (employee)

OU (Organizational Unit)

- Department/group (HR, IT, Finance)

GPO (Group Policy Object)

- Rules applied to OU

Domain Controller

- Machine where AD is installed
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⚙️ How AD Works (Real Flow)

1. Admin creates user 👤
 2. Sets username & password 🔑
 3. Manager shares credentials
 4. User logs in → forced to change password 🔁
 5. GPO automatically applied 🎯
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🔄 If Policy Not Applied?

- 👉 Run Command Prompt (Admin):
gpupdate /force
 - 👉 Forces system to connect with Domain Controller
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🔗 4. Domain Join Methods

✅ Method 1: System Properties

1. Right-click **This PC** → **Properties**
 2. Click **Change Settings**
 3. Click **Domain**
 4. Enter domain name
 5. Enter admin credentials
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✅ Method 2: Settings

1. Go to:

Settings → Accounts → Access Work or School

2. Click **Join this device to a domain**

3. Enter domain ID

Requirements:

- Same network (Intranet) 
 - OR VPN if remote 
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5. VPN Types

◆ Site-to-Site VPN

- Connects **office to office**
- Example:
 - Mumbai ↔ Pune

 Secure tunnel between routers

◆ Remote Access VPN

- Connects **user to office**

 Tools:

- Cisco AnyConnect
 - GlobalProtect
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6. PRACTICAL TASK – FULL STEP-BY-STEP

PART 1: Install Active Directory

1. Open **Server Manager**
 2. Click **Add Roles and Features**
 3. Select:
 - Role-based installation
 -  **Active Directory Domain Services (AD DS)**
 4. Click **Install**
 5. After install → Click:
 **Promote this server to Domain Controller**
 6. Select:
 - **Add new forest**
 - Domain: company.local
 7. Set password → Install → Restart
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PART 2: Create OUs

1. Open **Active Directory Users and Computers**
 2. Right-click company.local → **New** → **Organizational Unit**
 3. Create:
 - IT
 - HR
 - Finance
 - Marketing
-

✓ PART 3: Create Users

1. Right-click OU → **New** → **User**
2. Enter:
 - Name
 - Username
 - Password
3. Create:
 - 5 users in each OU (Total = 20)

✓ PART 4: Create GPO (Block Control Panel)

1. Open **Group Policy Management**
2. Right-click OU → **Create GPO**
 - Name: Block_Control_Panel
3. Right-click GPO → **Edit**

👉 Go to:

User Configuration

→ Policies

→ Administrative Templates

→ Control Panel

4. Enable:

👉 **Prohibit access to Control Panel and PC Settings**

✓ PART 5: Link GPO

- Right-click OU → **Link Existing GPO**
 - Select: Block_Control_Panel
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✓ PART 6: Join Client to Domain

Ensure:

- **Client machine is on same network (or NAT in VM)**
1. Open **System Properties** (sysdm.cpl)
 2. Click **Change**
 3. Select **Domain** → Enter: company.local
 4. Enter admin credentials
 5. Restart

✓ Verification:

Login with domain user:

- 👉 Try opening Control Panel
- ✗ It should be blocked

🎯 FINAL RESULT

You have successfully:

- Installed Active Directory
 - Created OUs
 - Created Users
 - Applied GPO
 - Joined client to domain
 - Blocked Control Panel
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Final Troubleshooting Tips

- ✓ If GPO not working → run gpupdate /force
 - ✓ Check network connectivity
 - ✓ Verify domain join
 - ✓ Restart system
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Final Summary

- BSOD = **System Protection Mechanism** 
 - Servers = **Service Providers** 
 - AD = **Centralized Control System** 
 - GPO = **Rules Engine** 
 - Domain = **Network Identity** 
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Q & A

SECTION 1: BSOD (Blue Screen of Death) – Interview Questions

 **Q1: What is BSOD and why does it occur?**

 **Answer:**

BSOD (Blue Screen of Death) is a **critical system error screen** in Windows that appears when the operating system encounters a **fatal error it cannot recover from**.

 It immediately stops the system to **prevent hardware damage or data corruption**.

 **Simple Explanation:**

Like an **MCB trip**  — system shuts down to protect itself.

 **Common Causes:**

- Faulty RAM 
- Driver issues 
- OS corruption 
- Disk errors 
- Overheating 
- Power supply issues 

 **Q2: How do you troubleshoot a BSOD issue step-by-step?**

 **Answer (Professional Approach):**

 **Step-by-Step Troubleshooting:**

 **Note Error Code**

- Example: IRQL_NOT_LESS_OR_EQUAL

- Helps identify root cause

2 Boot into Safe Mode

- To isolate driver/software issue

3 Check Recent Changes

- New driver or software install?

4 Run System File Check

sfc /scannow

5 Run Disk Check

chkdsk /f /r

6 Update Drivers

- Especially GPU, chipset, network

7 Check RAM

- Use Windows Memory Diagnostic

8 Check Event Viewer

- Analyze crash logs 📁

9 Check Temperature

- Overheating causes shutdown

10 Last Option

- System Restore or OS Repair

? Q3: What tools are used to analyze BSOD?

✓ Answer:

- Event Viewer 📁
- Windows Memory Diagnostic 🧠
- BlueScreenView 🔍

- WinDbg (Advanced debugging tool)

? Q4: What is the difference between hardware BSOD and software BSOD?

✓ Answer:

Type	Cause	Example
Hardware BSOD	Physical issue	Faulty RAM
Software BSOD	OS/driver issue	Bad driver

👉 Hardware = Replace component

👉 Software = Fix OS/driver

? Q5: What is MiniDump file?

✓ Answer:

MiniDump is a **small crash log file** created during BSOD.

📍 Location:

C:\Windows\Minidump

👉 Used for **debugging crash reasons**

💻 SECTION 2: SERVER – Interview Questions

? Q6: What is a server?

✓ Answer:

A server is a **computer or system that provides services to other computers (clients)** over a network.

🧠 Example:

- Laptop = Client 💻

- Server = Service provider 

? Q7: What are different types of servers?

✓ Answer:

Server Type	Function
Web Server 	Hosts websites
File Server 	Stores files
DNS Server 	Domain → IP
DHCP Server 	Assigns IP
Mail Server 	Email services
AD Server 	User management
Database Server 	Stores data

? Q8: What is the difference between client and server?

✓ Answer:

Client 	Server 
Requests service	Provides service
End user machine	Backend system

? Q9: What is DHCP?

✓ Answer:

DHCP (Dynamic Host Configuration Protocol) automatically assigns:

- IP address 
- Subnet mask

- Gateway

👉 Without DHCP → manual IP needed

? Q10: What is DNS?

✅ Answer:

DNS converts **domain names into IP addresses**

👉 Example:

google.com → 142.x.x.x

🏢 SECTION 3: ACTIVE DIRECTORY (AD) – Interview Questions

? Q11: What is Active Directory?

✅ Answer:

Active Directory is a **centralized directory service by Microsoft** used to manage:

- Users 👤
 - Computers 💻
 - Policies 🔒
 - Resources 📁
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? Q12: What is OU (Organizational Unit)?

✅ Answer:

OU is a **container inside Active Directory** used to organize:

- Departments
- Users
- Systems

👉 Helps in applying policies easily

? Q13: What is Group Policy (GPO)?

✓ Answer:

GPO is used to **apply rules/settings to users or computers centrally**.

🎯 Example:

- Disable Control Panel 🚫
 - Set company wallpaper 🖼️
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? Q14: What is a Domain Controller?

✓ Answer:

A Domain Controller is a **server that runs Active Directory services** and handles:

- Authentication 🗝️
 - Authorization 🚫
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? Q15: What is Authentication vs Authorization?

✓ Answer:

Concept	Meaning
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Authentication	Who are you? 🗝️
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Authorization	What can you access? 🚫
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? Q16: What is Domain Join?

✓ Answer:

Connecting a computer to a domain so it can:

- Follow company policies
 - Use centralized login
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? Q17: What is gpupdate /force?

✓ Answer:

Command used to **force apply Group Policy immediately**

gpupdate /force

⚙ SECTION 4: PRACTICAL / JOB-ORIENTED QUESTIONS

? Q18: GPO is not applying. What will you do?

✓ Answer (Stepwise):

- 1 Check system is domain joined
- 2 Check OU is correct
- 3 Verify GPO is linked
- 4 Run:

gpupdate /force

- 5 Run:

gpresult /r

- 6 Restart system
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? Q19: User cannot login to domain. How will you fix?

✓ Answer:

- 1 Check network connectivity 🌐
 - 2 Verify domain server reachable
 - 3 Check user account status
 - 4 Reset password 🔑
 - 5 Check time sync 🕒
 - 6 Check DNS configuration
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? Q20: How do you create a user in AD?

✓ Answer:

- 1 Open AD Users & Computers
 - 2 Right-click OU
 - 3 Click → New → User
 - 4 Enter details
 - 5 Set password
 - 6 Enable account
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? Q21: What is onboarding and offboarding?

✓ Answer:

Process	Meaning
Onboarding 🧑	New user creation
Offboarding 🚪	Disable/delete user

? Q22: How do you block Control Panel using GPO?

✓ Answer:

1 Open Group Policy Management

2 Create GPO

3 Go to:

User Config → Admin Templates → Control Panel

4 Enable:

👉 Prohibit access

? Q23: What are prerequisites for domain join?

✓ Answer:

- Same network 🌐
 - Proper DNS
 - Domain credentials 🔒
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🔒 SECTION 5: SECURITY & REAL SCENARIOS

? Q24: Why should admin rights be limited?

✓ Answer:

- Prevent malware spread 🦠
 - Reduce risk of system damage
 - Improve security 🔒
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? Q25: What happens if GPO is misconfigured?

✓ Answer:

- Users lose access
 - System restrictions applied wrongly
 - Security issues
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? Q26: How does AD improve security?

✓ Answer:

- Central control
 - Policy enforcement
 - Access restriction
 - Logging & auditing
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FINAL SUMMARY

Concept	Meaning
BSOD 	System crash protection
Server 	Service provider
AD 	Central management
OU 	Grouping system
GPO 	Rule engine
Domain 	Network identity
